

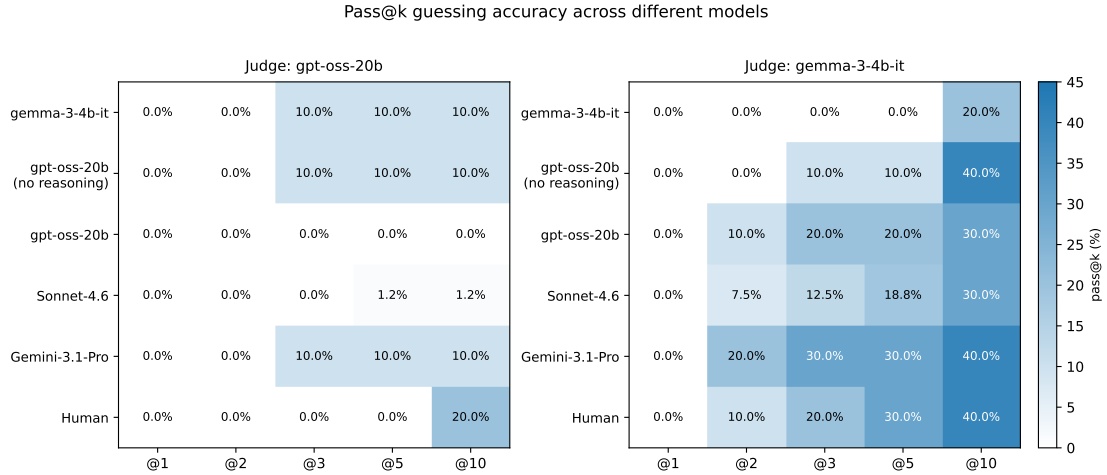


Figure 4: Pass@ k guessing accuracy on 10 shared Taboo cards, for all clue-givers evaluated by openai/gpt-oss-20b (left) and google/gemma-3-4b-it (right). Each cell reports the fraction of cards for which the judge ranked the target word within its top k guesses.

is 0.0% while pass@10 reaches 30.0%. The same asymmetry holds when models guess from model-written descriptions, with Claude Sonnet-4.6 and Gemini-3.1-Pro as clue givers yielding comparable pass@10 of 25.0% and 30.0% respectively under google/gemma-3-4b-it judge, while openai/gpt-oss-20b remains at or below 20.0% across all clue givers.

4.4. Humans versus models; models versus humans

Humans as guessers. We asked human annotators to read model-generated descriptions and guess the target word. Overall, annotators correctly identified the target in 30.77% of cases (weighted average across annotators and conditions), with performance varying considerably by constraining strategy: constrained generation yields the highest human accuracy (38.10%), followed by the SAE-based condition (31.03%), with prompting producing the lowest (23.44%). This ordering mirrors the pattern observed in Section 4.3 for model judges, suggesting that the informativeness advantage of constrained generation generalises beyond automatic evaluation.

Beyond aggregate accuracy, qualitative inspection reveals recurring failure patterns in model-generated descriptions, including referential failures, category mismatches, and code-switching under the SAE condition, which we discuss in Appendix D.

Humans as descriptors. We also collected human-generated descriptions across three conditions that differ in prior exposure and time constraints (see *description phase* in Section 3.3): *spontaneous* (no preparation, 5 cards), *deliberate* (time to reflect before writing, 5 cards), and *shared* (all annotators describe privately the same ten words, enabling direct comparison with models). For the *spontaneous* and *deliberate* conditions, we assessed how accurately other annotators could guess the target word from the description: *deliberate* descriptions achieve 85% accuracy, while *spontaneous* descriptions yield a higher rate of 95%. The gap between these

two reflects the interactive nature of oral description, where players could refine in real time, versus the deliberate setting, which more closely mirrors the constraints imposed on models.

If we ask the models to guess the human-described words, we can spot a clear ordering: shared > deliberate > spontaneous at all values of k (Table 1). At pass@10, shared descriptions yield 15.6%, deliberate 13.8%, and spontaneous only 5.1%. As observed throughout, openai/gpt-oss-20b almost never guesses correctly (pass@10 $\leq$ 2.5%), whereas google/gemma-3-4b-it reaches up to 30.0% on shared descriptions, yet still requiring between 5 and 10 guesses to rival what a human achieves at pass@1.

This contrast is revealing: humans benefit from spontaneous descriptions that, though fragmented, are iteratively refined towards the referent in real time. When presented to models as static text (without the interactive context that makes them effective for human guessers) such descriptions yield substantially lower pass@ k rates, while models are better suited to process the structured, fixed outputs of the deliberate and shared conditions.

Table 1

Human clue-giver evaluation. Accuracy computed on strict criterion (no exact + no morphological violations).

Condition	Judge	pass@ k					Accuracy	
		@1	@2	@3	@5	@10	loose	strict
Spontaneous ($n = 39$)	gpt-oss-20b	0.0%	0.0%	0.0%	0.0%	0.0%	97.4%	82.1%
	gemma-3-4b-it	0.0%	2.6%	5.1%	7.7%	10.3%		
Deliberate ($n = 40$)	gpt-oss-20b	0.0%	0.0%	0.0%	0.0%	2.5%	100.0%	97.5%
	gemma-3-4b-it	0.0%	5.0%	7.5%	10.0%	25.0%		
Shared ($n = 80$)	gpt-oss-20b	0.0%	0.0%	0.0%	1.2%	1.2%	100.0%	91.2%
	gemma-3-4b-it	0.0%	7.5%	12.5%	18.8%	30.0%		

5. Discussion

Our results reveal a consistent tension between constraint adherence and descriptive informativeness that cuts across all conditions. The three families of constraining strategies we examine operate at qualitatively different levels (instruction, decoding, and internal representation) and this difference determines not only how reliably the model avoids forbidden words, but also how it describes the target item. Prompting achieves high compliance at the cost of informationally sparse output; constrained generation forces the model toward more semantically precise paraphrases, at the cost of occasionally leaking the target word; SAE-based steering sits at the opposite extreme, preserving conceptual associations at the cost of surface compliance. This trade-off suggests that lexical avoidance and communicative effectiveness are not independently controllable, and that the mechanism through which a constraint is imposed shapes the output in ways that go beyond mere compliance.

A second thread concerns the asymmetry between humans and models as players. On the descriptor side, reasoning does help models: openai/gpt-oss-20b with chain-of-thought reaches 97.4% accuracy versus 72.2% without, a substantial gain. For humans, the picture is

more nuanced: spontaneous oral descriptions, despite being unplanned, yield higher guessing accuracy (95.0%) than deliberate written ones (85.0%), likely because the interactive oral setting allows real-time refinement. This suggests that planning confers different benefits depending on the medium and the player: for models it primarily enforces compliance, whereas for humans it is the interactivity of the setting, rather than deliberation *per se*, that drives performance. On the guesser side, the asymmetry is starker: both judge models struggle to identify target words, with `google/gemma-3-4b-it` requiring between 5 and 10 guesses to approach the accuracy a human achieves at `pass@1`. This gap suggests that models do not share the same network of salient lexical associations that makes Taboo intuitive for humans, and that guessing from indirect descriptions remains a genuinely hard task for current language models. This gap is also reminiscent of the distinction between fast, associative System 1 processing and deliberate System 2 reasoning [27]: humans navigate Taboo by rapidly activating and suppressing salient lexical associates, a mode of processing that current language models do not appear to replicate, at least not in the guessing direction.

Our study has however several limitations. The benchmark is currently Italian-only, and it is unclear how results generalise to languages with different morphological profiles or lexical structures. The shared evaluation set covers only 10 words, limiting the statistical power of cross-model comparisons in Section 4.3. SAE-based steering was evaluated on `google/gemma-3-4b-it` only, and extending it to other architectures remains open. The two judge models differ substantially in sensitivity, raising questions about which better reflects human guessing behaviour, a question our human evaluation begins to address but does not fully resolve.

6. Conclusions and Future Work

We presented a framework that operationalises lexical constraint following as a communicative game, and used it to evaluate three families of constraining strategies across multiple models and human annotators. Our results show that compliance and communicative effectiveness are not independently controllable: the mechanism through which a constraint is imposed (instruction, decoding, or internal representation) shapes the output in ways that go beyond mere rule adherence, with constrained generation and SAE-based steering producing qualitatively different trade-offs. Beyond the specific findings, our work supports the broader argument that games constitute a natural and underexplored testbed for language model capabilities: they impose well-defined rules, require communicative grounding, and admit quantitative evaluation while remaining ecologically valid. LMtaboo in particular is inherently interactive, yet we have evaluated models only in a single-turn, single-agent setting. Extending the benchmark to multi-turn and multi-agent configurations (where models iteratively refine descriptions based on guesser feedback, or negotiate meaning across turns) would more faithfully reflect the dynamics of the original game and probe capabilities, such as pragmatic adaptation and collaborative grounding, that current evaluations leave largely untested.

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Declaration on Generative AI

During the preparation of this work, the authors used **Claude** (Anthropic) in order to **improve writing style, grammar and spelling check, paraphrase and reword, drafting content** (including experimental code). After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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